

Geometry: Trapezoid Midsegment & Kite



Practice Worksheet • numberbender.com

Name: _____

Date: _____

Score: / 10

DIRECTIONS

Apply the Midsegment (Midline) Theorem and kite properties to solve each problem. Show all work.

- 1 Find the midsegment m of the trapezoid:

$$b_1 = 9, \quad b_2 = 13$$

Answer: _____

- 2 Find the midsegment m of the trapezoid:

$$b_1 = 4, \quad b_2 = 18$$

Answer: _____

- 3 Find the missing base b_2 of the trapezoid:

$$m = 9, \quad b_1 = 6$$

Answer: _____

- 4 Find the missing base b_1 of the trapezoid:

$$m = 7, \quad b_2 = 9$$

Answer: _____

- 5 Find x using the Midsegment Theorem:

$$b_1 = 3x + 2, \quad b_2 = 3x + 6, \quad m = 10$$

Answer: _____

- 6 Find the non-vertex angles B and D of the kite:

$$\angle A = 70^\circ, \quad \angle C = 90^\circ$$

Answer: _____

- 7 Find the missing vertex angle A of the kite:

$$\angle B = \angle D = 90^\circ, \quad \angle C = 50^\circ$$

Answer: _____

- 8 Find the perimeter of the kite (two pairs of equal sides):

$$a = 11, \quad b = 17$$

Answer: _____

- 9 Diagonal d_2 is bisected by d_1 in the kite. Find the full length of d_2 :

$$\overline{WX} = 4$$

Answer: _____

- 10 Diagonals of a kite are perpendicular. Find angle x :

$$\angle 1 = 60^\circ, \quad \angle 2 = 90^\circ$$

Answer: _____



Answer Key & Solutions

Geometry: Trapezoid Midsegment & Kite • Numberbender



TEACHER NOTES

Midsegment: $m = (b_1 + b_2)/2$. Kite: non-vertex angles are equal; diagonals are perpendicular; one diagonal bisects the other. Angle sum = 360 degrees.

1 Find the midsegment m of the trapezoid:

$$b_1 = 9, \quad b_2 = 13$$

$$= m = 11$$

$$m = (b_1 + b_2)/2 = (9 + 13)/2 = 11.$$

2 Find the midsegment m of the trapezoid:

$$b_1 = 4, \quad b_2 = 18$$

$$= m = 11$$

$$m = (b_1 + b_2)/2 = (4 + 18)/2 = 11.$$

3 Find the missing base b_2 of the trapezoid:

$$m = 9, \quad b_1 = 6$$

$$= b_2 = 12$$

$$b_2 = 2m - b_1 = 2(9) - 6 = 12.$$

4 Find the missing base b_1 of the trapezoid:

$$m = 7, \quad b_2 = 9$$

$$= b_1 = 5$$

$$b_1 = 2m - b_2 = 2(7) - 9 = 5.$$

5 Find x using the Midsegment Theorem:

$$b_1 = 3x + 2, \quad b_2 = 3x + 6, \quad m = 10$$

$$= x = 2$$

$$\text{Solve } (3x + 2 + 3x + 6)/2 = 10 \text{ for } x = 2.$$

6 Find the non-vertex angles B and D of the kite:

$$\angle A = 70^\circ, \quad \angle C = 90^\circ$$

$$= \angle B = \angle D = 100^\circ$$

$$\text{Non-vertex angles are equal. } B = D = (360 - 70 - 90)/2 = 100 \text{ deg.}$$

7 Find the missing vertex angle A of the kite:

$$\angle B = \angle D = 90^\circ, \quad \angle C = 50^\circ$$

$$= \angle A = 130^\circ$$

$$A = 360 - 2(90) - 50 = 130 \text{ degrees.}$$

8 Find the perimeter of the kite (two pairs of equal sides):

$$a = 11, \quad b = 17$$

$$= P = 56$$

$$P = 2(11) + 2(17) = 22 + 34 = 56.$$

9 Diagonal d_2 is bisected by d_1 in the kite. Find the full length of d_2 :

$$\overline{WX} = 4$$

$$= \overline{WY} = 8$$

$$WY = 2 \times WX = 2 \times 4 = 8.$$

10 Diagonals of a kite are perpendicular. Find angle x :

$$\angle 1 = 60^\circ, \quad \angle 2 = 90^\circ$$

$$= \angle x = 30^\circ$$

$$\text{Angles in triangle sum to 180. } x = 180 - 90 - 60 = 30 \text{ deg.}$$

